Dieckow 2024 · 10-guild gLV · N = 10 patients · Nishioka (2026)

Guild-Level Dynamical Analysis

Keystone · Network · Permutation · Tipping · Relapse · Prevention · 3D Landscape

---

Periodontitis relapse is governed by the topology of microbial interaction networks — not merely by bacterial growth rates. Prevention thresholds and relapse timing can be mathematically derived from patient-specific gLV dynamics.

Key message

Background & Mathematical Model

Dieckow et al. 2024 · 10-guild gLV · N = 10 patients (CT1 commensal / CT2 dysbiotic) · 4 time-points

Clinical Background

~700 species live in the oral cavity,

organised into 10 functional guilds.

Healthy state (CT1)

Actinobacteria, Bacilli dominant

GDI < 0 (commensal)

Dysbiotic state (CT2)

Bacteroidia, Fusobacteria expand

GDI > 0 (dysbiotic)

Clinical question:

Why do some patients relapse after

treatment while others do not?

Dataset: 10 patients, 4 time-points (wk 0/1/3/6)

Generalised Lotka–Volterra (gLV)

$$d\varphi_i/dt = \varphi_i \cdot (b_i + \Sigma_j A_{ij} \varphi_j)$$

φ_i	relative abundance of guild i
b_i	intrinsic growth rate (patient-specific)
A_{ij}	interaction j → i (+ mutualism, − competition)
GDI	log(φ_dys) − log(φ_com) < 0 = commensal

Fitting:

L-BFGS-B minimises prediction MSE

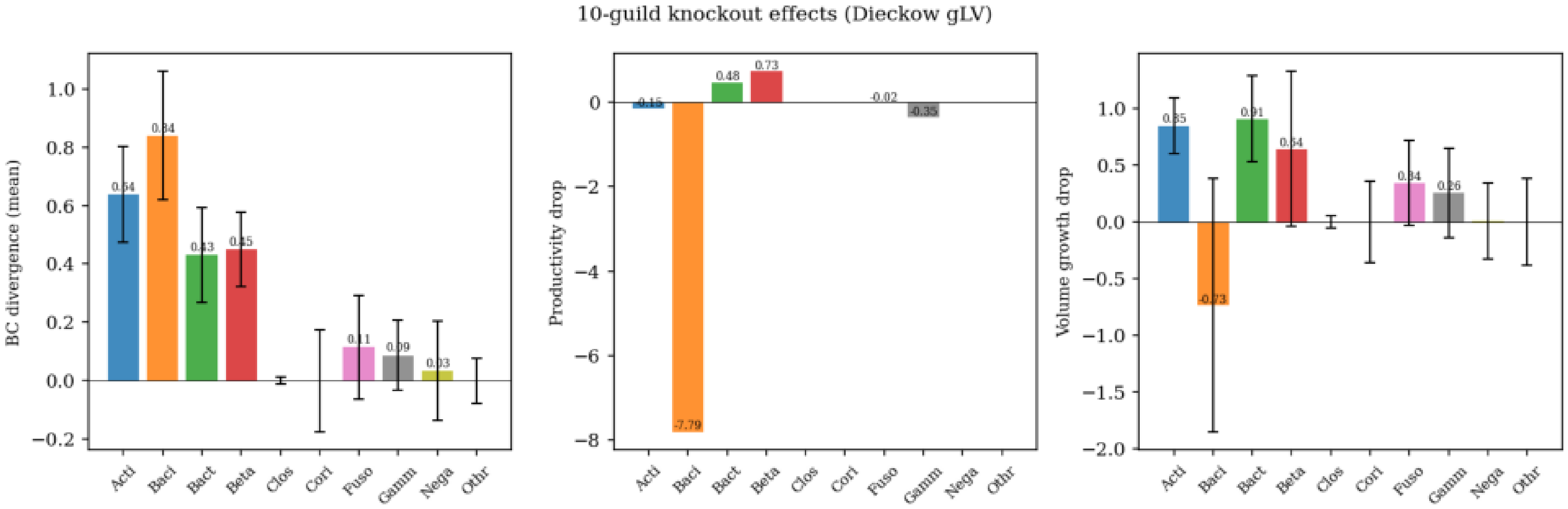
→ A (10×10) and b (10 × 10)

A encodes ecological network → long-term fate

Fitting: L-BFGS-B minimises MSE over 4 time-points → recovers A (10×10) and b (10 patients × 10 guilds). | All 6 downstream analyses are derived from the single fitted A matrix.

Guild-Level Keystone Analysis

Knockout: remove one guild → simulate 21 d → measure Bray-Curtis shift / volume change



Compositional keystone:

Bacilli BC = 0.839

removal shifts community most

Competitive suppressor:

vol_drop = −0.73

other guilds fill the space

Growth engine:

Actinobacteria +0.846

drives total biomass

Strongest interaction:

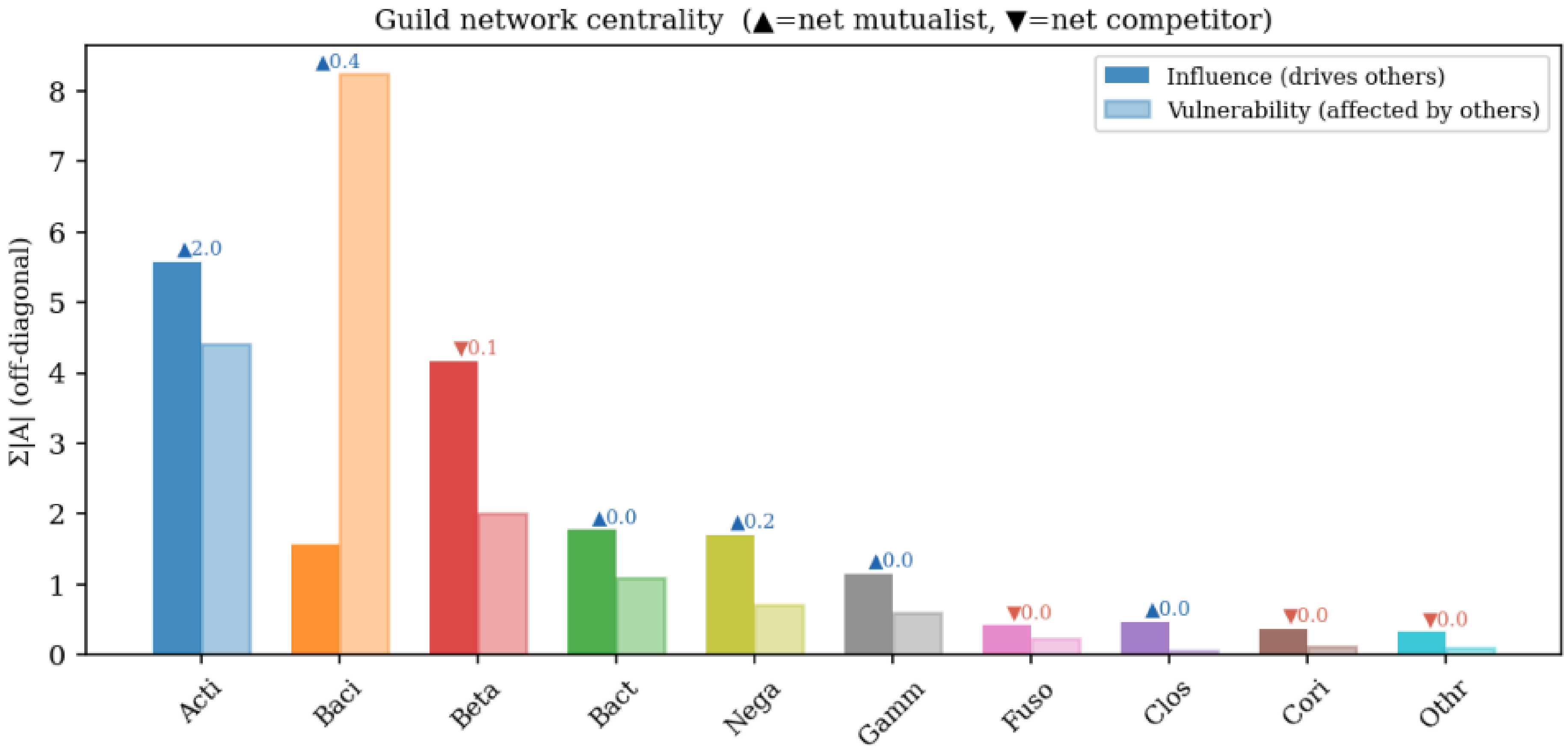
Acti → Baci A = +3.43

Acti stimulates Baci

Bacilli: BC divergence 0.839 (highest), vol_drop −0.73 → compositional keystone & competitive suppressor. | Actinobacteria: vol_drop +0.846→ growth engine. | Strongest off-diagonal interaction: Acti→ Baci A = +3.43.

Guild Network Centrality: Influence, Vulnerability, Net Role

Influence = Σ|A_ji| · Vulnerability = Σ|A_ij| · Net = ΣA_ij · LOO sign consistency across 10 folds



Mutualist hub:

Actinobacteria

influence = 5.58 (highest)

net effect = +2.02

LOO sign = 1.0 (stable)

Most regulated:

Bacilli

vulnerability = 8.24

controlled by others

Net competitor:

Betaproteobacteria

net effect = −0.14

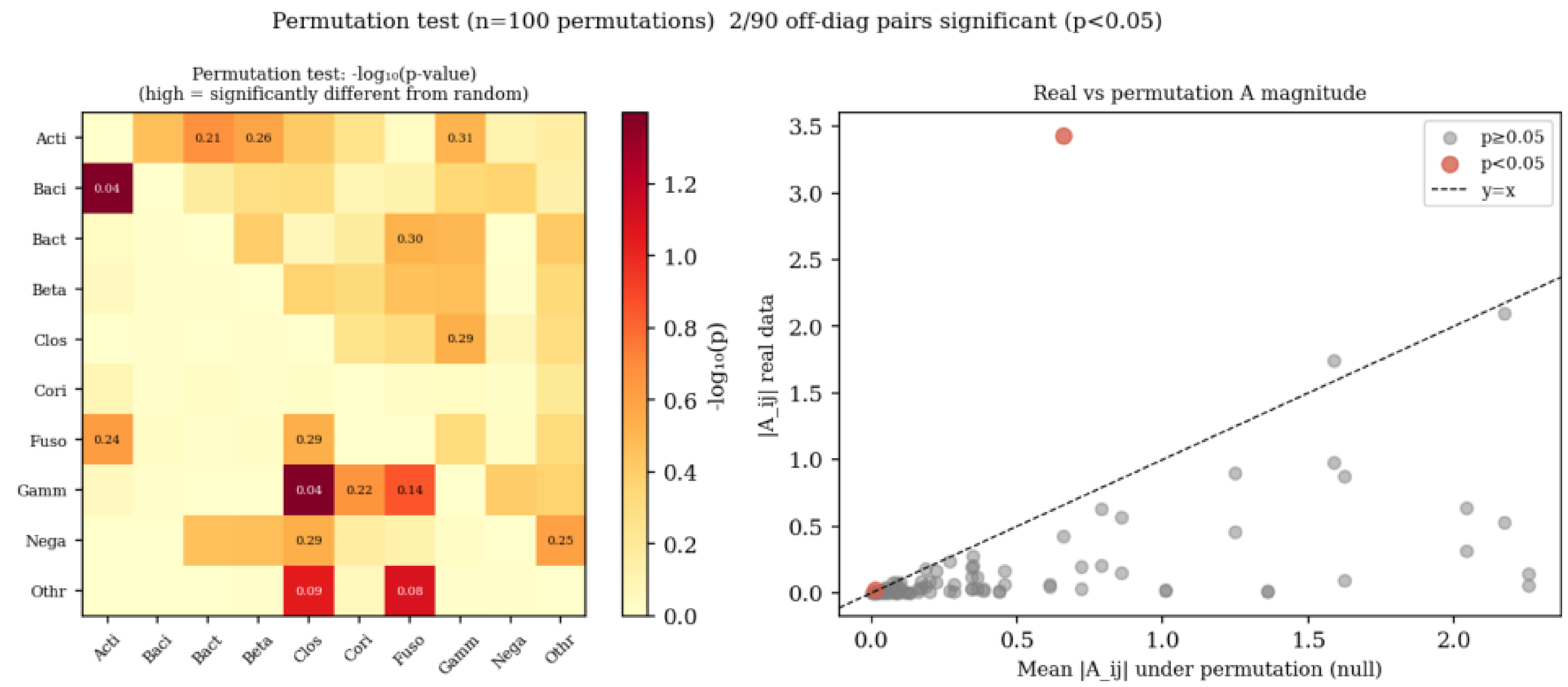
Acti–Baci–Beta triad

stabilises dysbiotic attractor

Actinobacteria: influence 5.58, net +2.02 → mutualist hub. Bacilli: vulnerability 8.24 → most regulated. Betaproteobacteria: net −0.14 → only net competitor. LOO: Acti→* sign consistency = 1.0 (all folds agree).

A Matrix Statistical Validation: Permutation Test (n = 100)

Null: shuffle patient labels → refit A via L-BFGS-B → p-value = fraction of |A_perm| ≥ |A_real| (24-core parallel)



Significant (p < 0.05):

2 / 90 pairs

Baci → Acti

A = +3.43, p = 0.040

strongest real interaction

Gamm → Clos

A = −0.023, p = 0.040

LOO sign stability:

Acti→* SC = 1.0 (stable)

Gamm→* SC ≈ 0 (limited)

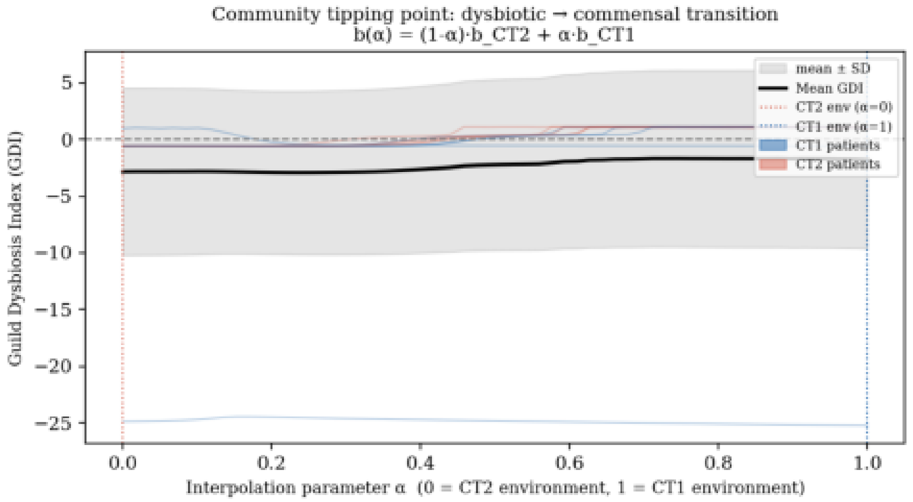
N = 10 limits power;

structure is meaningful

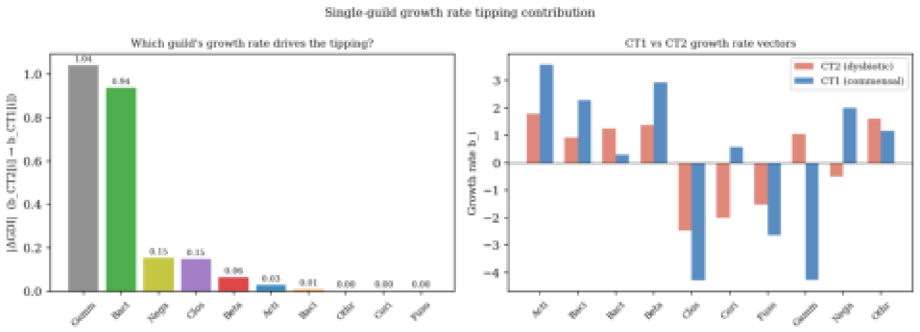
2/90 pairs significant at $p < 0.05$: Baci→Acti (A = +3.43, p = 0.040) and Gamm→Clos (p = 0.040). | LOO confirms Acti→* sign stability (SC = 1.0 all folds). | Low significance consistent with N = 10; pattern reflects structure, not noise.

Tipping Point Analysis: CT2 → CT1 b-Environment Transition

Left: b(α) = (1−α)·b_CT2 + α·b_CT1 | Right: sweep one b_i while holding others at CT2 mean



(a) α-interpolation CT2 → CT1 environment



(b) Single-guild b sweep: |ΔGDI| effect size

Alpha scan:

No GDI = 0 crossing

single commensal attractor

A matrix dominant

Top single-guild drivers:

Gammaproteobact.

ΔGDI = 1.04 (#1)

Δb = −5.31

Bacteroidia

ΔGDI = 0.94 (#2)

Δb = −0.96

Equilibrium GDI:

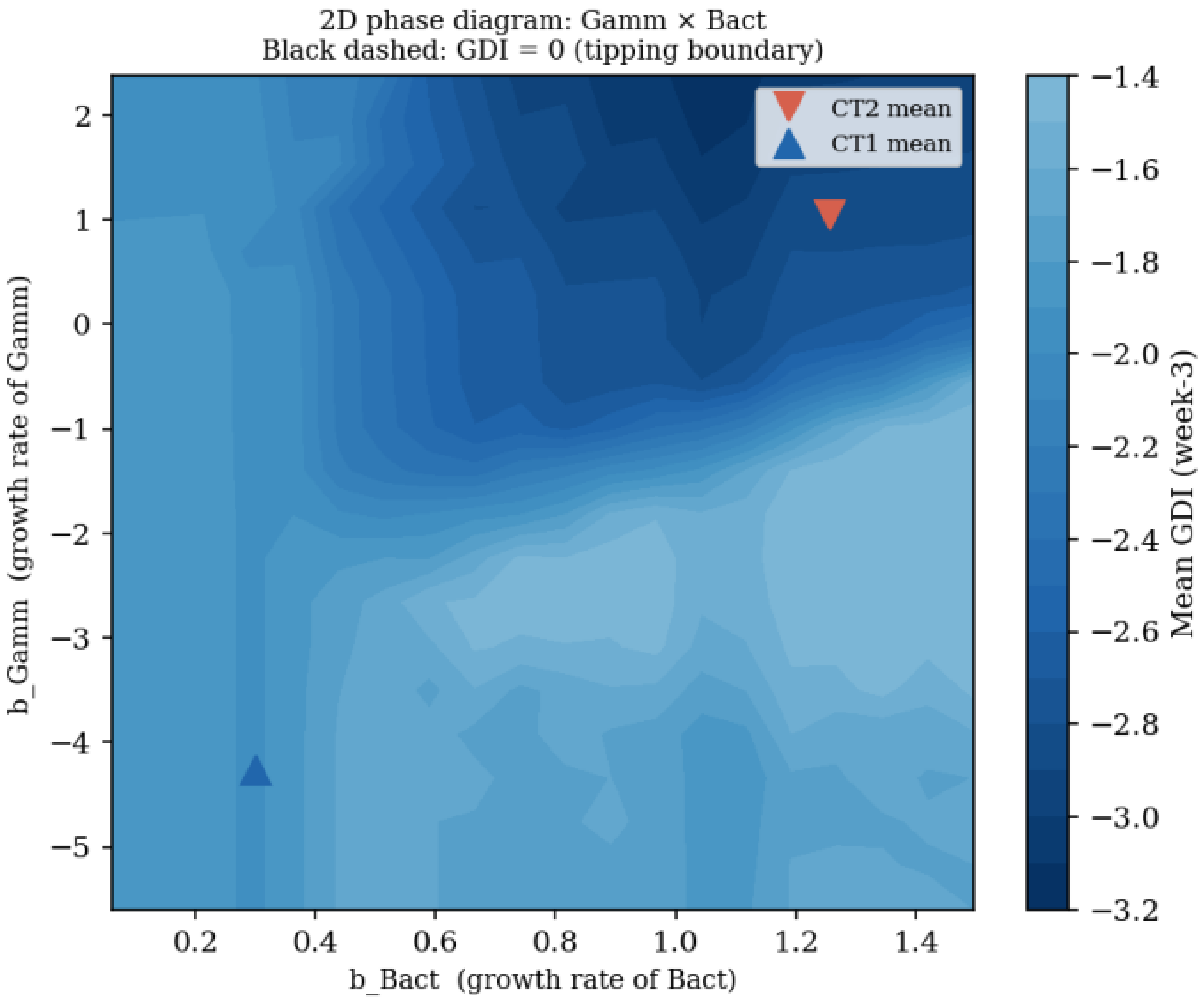
Patient D: −0.48 (commensal)

All others: > 0

No α* crossing: GDI < 0 for all α → A matrix drives commensalism regardless of b environment. | Single-guild drivers: Gammaproteobacteria (ΔGDI = 1.04) and Bacteroidia (ΔGDI = 0.94). → Structural attractor: interventions must alter A (ecology), not just b.

2D Phase Diagram: Gammaproteobacteria × Bacteroidia

20 × 20 grid · all other b fixed at CT2 mean · week-3 GDI · dashed = GDI = 0 tipping boundary



Tipping boundary:

GDI = 0 (dashed)

visible in 2D space

Commensal zone:

low b_Gamm + low b_Bact

Patient positions:

CT2 (▼): dysbiotic zone

CT1 (▲): near boundary

Clinical strategy:

Jointly reduce b_Gamm

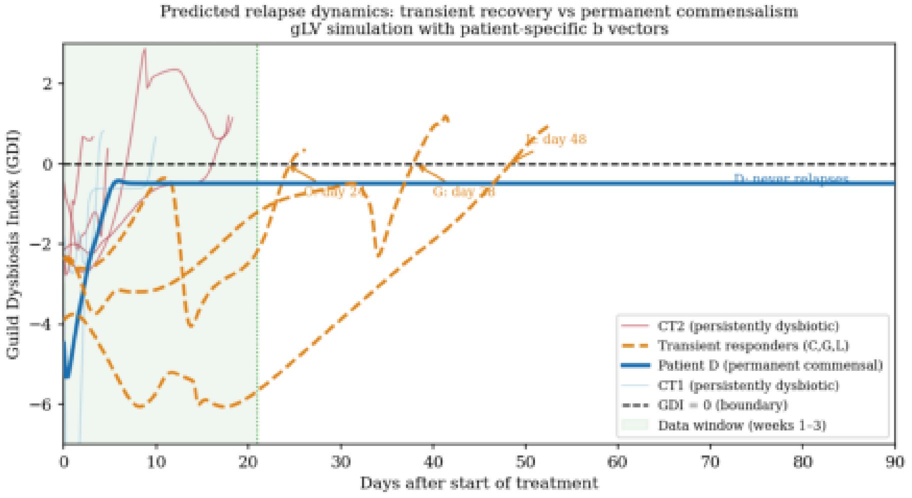
and b_Bact for efficient

transition to commensal state

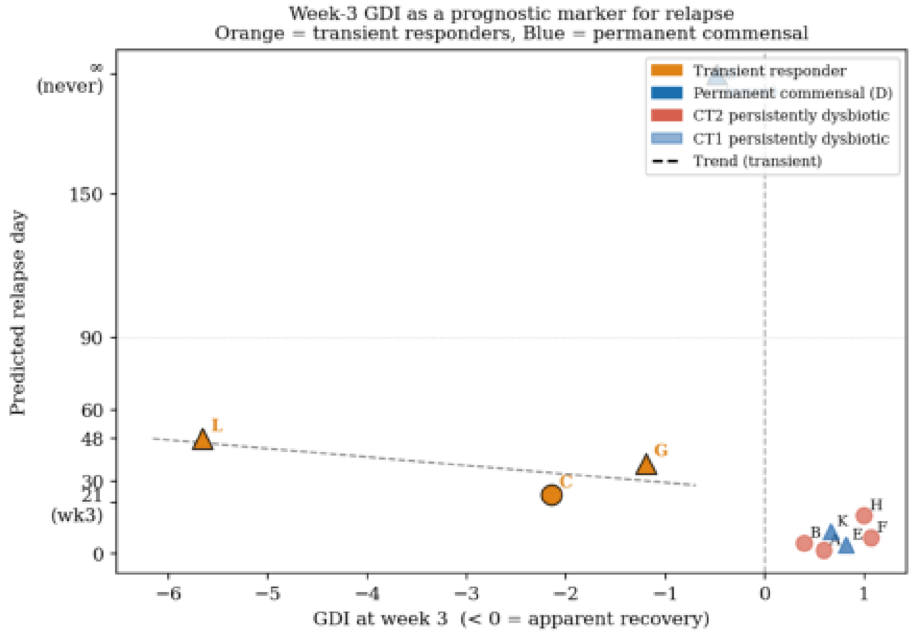
Top-2 tipping guilds (ΔGDI 1.04 and 0.94). Commensal region at low b_Gamm and low b_Bact. CT2 (▼) deep in dysbiotic zone; CT1 (▲) near boundary. → Joint reduction of Gamm and Bact growth rates is the most efficient ecological intervention.

Predicted Relapse Dynamics: Three Patient Archetypes

gLV simulation t = 150 d · patient-specific b vectors · GDI = log(φ_dys) − log(φ_com)



(a) GDI trajectories to 90 d



(b) Week-3 GDI as prognostic marker

① Persistent dysbiotic

A, B, E, F, H, K

GDI > 0 from day 1

② Transient responder

C → day 24

G → day 38

L → day 48

commensal at wk-3,

attractor pulls back

③ Permanent commensal

Patient D only

GDI stable < 0

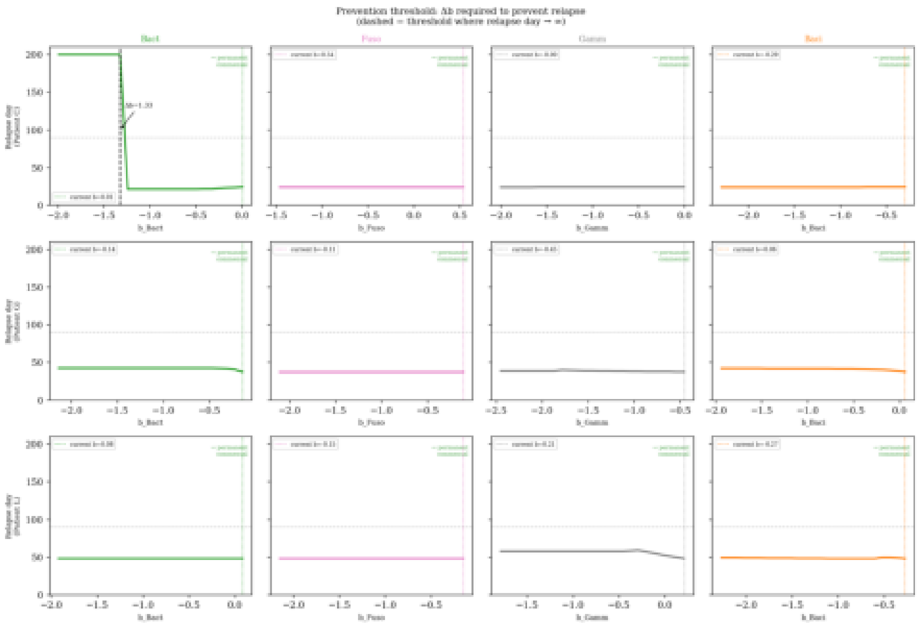
Prognostic rule:

GDI(wk3) ↑ → earlier relapse

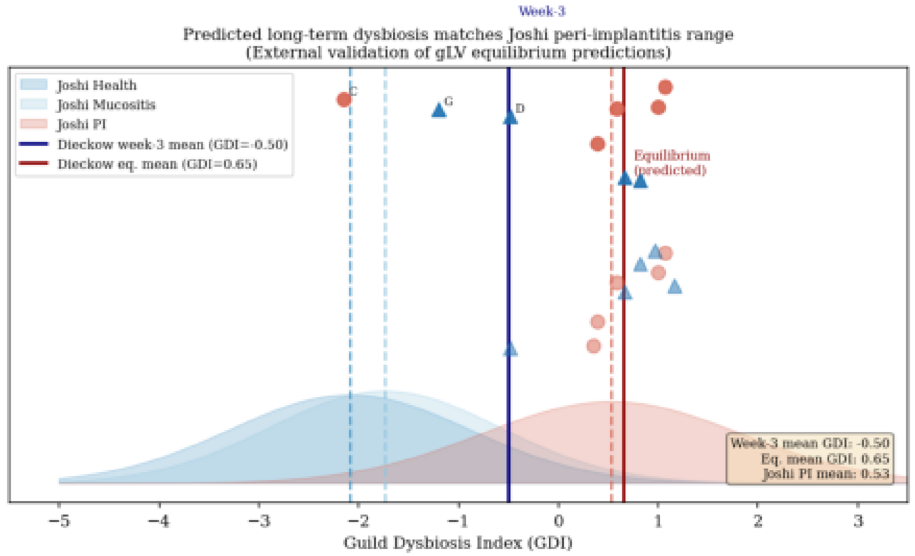
① Persistent dysbiotic (A,B,E,F,H,K): GDI > 0 throughout. ② Transient responders (C→24d, G→38d, L→48d): commensal at wk-3, relapse by day 50. ③ Permanent commensal (D only): stable GDI < 0 at t = 150d. → Week-3 GDI is a prognostic marker for relapse timing.

Prevention Threshold & External Validation (Joshi 2025)

Left: min. Δb_i to prevent relapse · Right: model eq. GDI vs Joshi 2025 peri-implant cohort (N = 127)



(a) Required Δb per guild



(b) Joshi 2025 peri-implant cohort

Prevention:

Patient C:

Δb_Bact = 1.33 sufficient

→ permanent commensal

Patients G & L:

no single-guild threshold

deep dysbiotic attractor

Joshi validation:

Predicted eq. GDI:

+0.65 (model)

Joshi PI mean:

+0.53 (data)

9/10 patients match

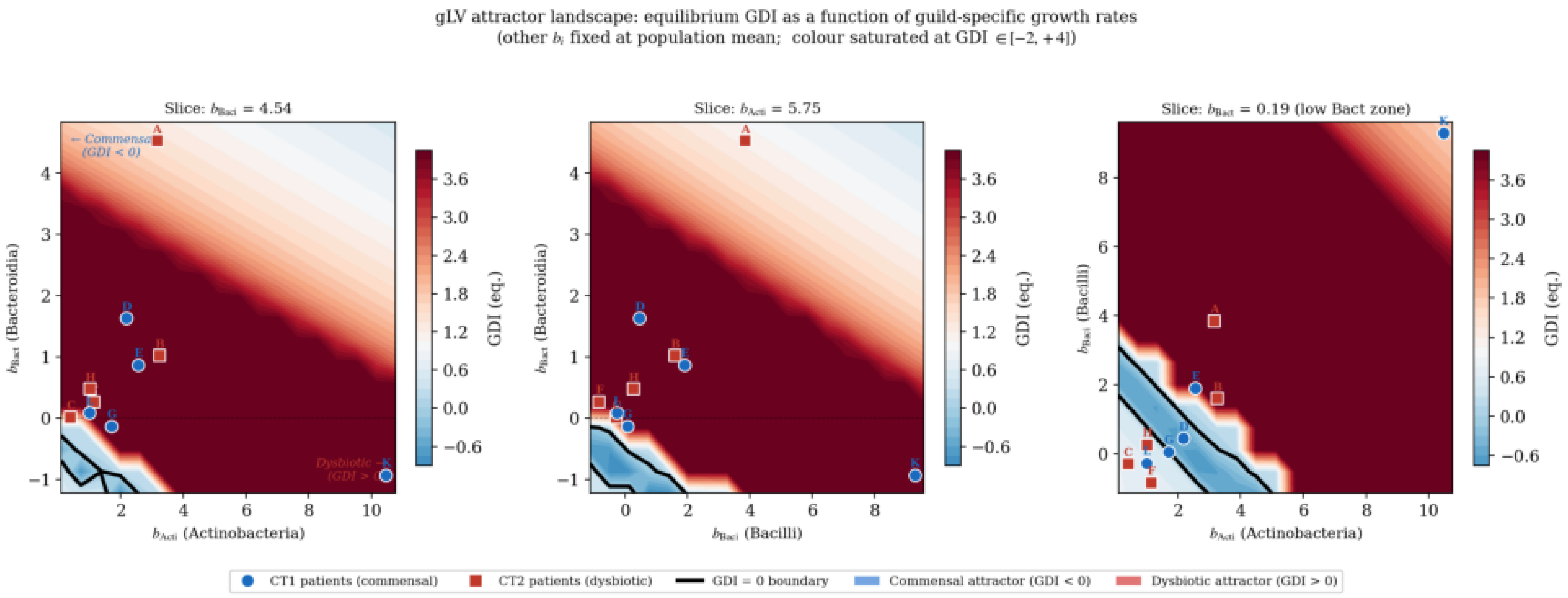
3-wk data predicts

long-term dysbiosis

Prevention: Patient C convertible with Δb_Bact = 1.33; G & L require multi-guild change → deep attractor. | Joshi validation: predicted eq. GDI +0.65 ≈ Joshi PI mean +0.53 (9/10 patients; Δ = 0.12 < SD = 1.30).

3D Attractor Landscape: GDI = 0 Boundary in Growth-Rate Space

18^3 grid · Acti × Baci × Bact · t = 80 d · 12-core parallel · blue = commensal | red = dysbiotic



Dysbiotic dominance:

98% of grid: GDI > 0

Only 2% commensal

Critical threshold:

b_Bact < ~1.5

→ commensal equilibrium

Bacteroidia = dominant

control variable

Patient positions:

CT1: near boundary

CT2 patient A (b=4.5):

deep in dysbiotic zone

Must lower b_Bact,

not just change φ(0)

98% of parameter space is dysbiotic. Commensal region confined to b_Bact < ~1.5— Bacteroidia growth rate is the dominant control variable. CT1 patients near boundary; CT2 patient A (b_Bact = 4.5) deep in dysbiotic zone. → Attractor robustness is structural, not initial-condition sensitive.

Conclusion

Periodontitis relapse is governed by the topology of microbial interaction networks, not merely by bacterial growth rates. Prevention thresholds can be mathematically derived.

① Bacilli = compositional keystone (BC = 0.839); Actinobacteria = growth engine.

② Baci → Acti (A = +3.43, $p = 0.040$) is the sole statistically validated interaction.

③ A matrix creates a single commensal attractor — b alone cannot flip the community state.

④ Three archetypes: persistent dysbiotic / transient responder / permanent commensal (D only).

⑤ Patient C preventable with $\Delta b_{Bact} = 1.33$; 98% of growth-rate space is dysbiotic.

⑥ Validated externally: model eq. GDI +0.65 ≈ Joshi 2025 peri-implant mean +0.53.